Module 3: Probability Rules — Practice

Lesson 1 | Calculate, justify, and interpret

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Preparation

Read Lesson 1. Use uniform selection of one record from the checked Restaurant CSV. Let A mean Very Good and B mean a price from USD 20 to less than USD 30. The full business introduction is in the lesson.

```
# read.csv() imports the checked teaching file; the analysis leaves its observations
 ↪  unchanged.
# >=20 and <30 define a half-open price band, so USD 30 is excluded.
r <- read.csv("data/module02/restaurant_ratings.csv")
A <- r$quality_rating == "Very Good"
B <- r$meal_price_usd >= 20 & r$meal_price_usd < 30
```

1. Define the experiment

State the sample space, one outcome, and the events A and B. Explain why restaurant identifiers are categorical labels even though R may import them as numbers. Are the three quality categories equally likely in this experiment?

2. Count event regions

Find the counts for both conditions, only A, only B, and neither condition. Check that they total 300. Create a labeled four-cell table or chart.

3. Calculate and interpret

Calculate $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, $P(A^c)$, and the probability of **exactly one** of A and B. Show the denominator and interpret the difference between “or” and “exactly one.”

4. Explain two errors

1. A student adds $P(A) + P(B)$ to calculate the union. Identify the double-counted records and correct the result.
2. Another student says the result proves the proportion of all current Los Angeles restaurants with those characteristics. Explain the limitation.

5. Count pairs

How many ordered pairs of distinct records are possible without replacement? How many unordered pairs? Explain why a count of possibilities alone does not identify an event probability.

Submit a short Quarto report with the calculations, one display, and interpretations. Return to the lesson.